# Property Valuation — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 21-March-2026

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# 1 Test Results — Property Valuation (MKM-PV-001)

Tests run on **21 March 2026 at 13:30:41**.

Table 1: Test Results Summary — Property Valuation (MKM-PV-001)

| | |
|---|---|
| Total Tests | 55 |
| Passed | 55 |
| Failed | 0 |
| Skipped | 0 |
| Duration | 16.55s |

Table 2: Detailed Test Results — Property Valuation

| Test Description | Acceptance Criteria | Result | Time |
|---|---|---|---|
| Line 541: non-dict schema → returns {}. | `result == {}` | Pass | 0.000s |
| Line 544-545: 'type'/'options'/'description' fields are skipped. | `"type" not in result; "options" not in result` | Pass | 0.000s |
| Calculate insurance premium flood risk high | `premium > 0` | Pass | 0.088s |
| Line 214: area=None → computes it first. | `isinstance(premium, float); premium > 0` | Pass | 0.165s |
| Line 211: property_value=None → computes it first. | `isinstance(premium, float); premium > 0` | Pass | 0.173s |
| Calculate monthly rent all types | `800 <= rent <= 15_000` | Pass | 0.000s |
| Lines 165-166: area=None → computes it first. | `isinstance(rent, float); rent > 0` | Pass | 0.000s |
| Lines 162-163: property_value=None → computes it first. | `isinstance(rent, float); rent > 0` | Pass | 0.087s |
| Calculate monthly rent positive | `isinstance(rent, float); 800 <= rent <= 15_000` | Pass | 0.000s |
| Calculate property area all types | `area > 0` | Pass | 0.000s |
| Lines 128-129: invalid date format → random years_ago. | `isinstance(price, float)` | Pass | 0.000s |
| Lines 120-121: property_value=None → computes it first. | `isinstance(price, float); price > 0` | Pass | 0.085s |
| Calculate sale price positive | `isinstance(price, float); 50_000 <= price <= 8_000_000` | Pass | 0.000s |

| Test Description | Acceptance Criteria | Result | Time |
|---|---|---|---|
| Lines 135-137: sale_date 1-2 years ago. | `isinstance(price, float)` | Pass | 0.000s |
| Lines 139-140: sale_date ¿ 2 years ago. | `isinstance(price, float)` | Pass | 0.000s |
| Lines 124-128: sale_date within 1 year. | `isinstance(price, float)` | Pass | 0.000s |
| Line 60-61: datetime → isoformat string. | `"2024-01-01" in result` | Pass | 0.000s |
| Lines 64-65: np.floating → float. | `"3.14" in result` | Pass | 0.000s |
| Lines 62-63: np.integer → int. | `result == "42"` | Pass | 0.000s |
| Lines 66-67: np.ndarray → list. | `result == [1, 2, 3]` | Pass | 0.000s |
| Line 68: super().default() raises TypeError for unknown type. | — | Pass | 0.000s |
| Lines 510-517: each fallback location has lat/lon/name/elevation. | `"lat" in loc`; `"lon" in loc` (+2 more) | Pass | 0.000s |
| Lines 499-518: fallback generates right number of locations. | `len(locs) == 5` | Pass | 0.001s |
| Lines 505-506: uses params.get_elevation if available. | `all(loc["elevation"] == 10.0 for loc in locs)` | Pass | 0.000s |
| Lines 608-612: generate_properties returns result dict. | `isinstance(result, dict)`; `"data" in result` (+1 more) | Pass | 0.780s |
| Lines 610-612: generate_properties creates output file. | `result["file_path"].exists()` | Pass | 0.528s |
| Calculate mortgage financials returns dict | `isinstance(result, dict)`; `"loan_amount" in result` (+2 more) | Pass | 0.000s |
| Generate financial data keys | `"mortgage_id" in data`; `"loan_amount" in data` (+1 more) | Pass | 0.000s |
| Interest rate range | `0.01 <= data["interest_rate"] <= 0.15` | Pass | 0.000s |
| Loan amount less than value | `data["loan_amount"] <= info["property_value"] * 1.05` | Pass | 0.000s |
| Mortgage id format | `re.match(r"^MORT-[a-f0-9]{8}$", data["mortgage_id"])` | Pass | 0.000s |
| Quality consistency check | `isinstance(corrected, dict)` | Pass | 0.000s |
| Calculate insurance premium positive | `isinstance(premium, (int, float))`; `premium > 0` | Pass | 0.087s |
| Calculate property area positive | `isinstance(area, (int, float))`; `area > 0` | Pass | 0.000s |
| Calculate property value in range | `50_000 <= value <= 10_000_000` | Pass | 0.877s |
| Generate bedrooms detached | `isinstance(beds, int)`; `1 <= beds <= 7` | Pass | 0.000s |
| Generate bedrooms flat | `isinstance(beds, int)`; `1 <= beds <= 4` | Pass | 0.000s |

| Test Description | Acceptance Criteria | Result | Time |
|---|---|---|---|
| Generate postcode format | `isinstance(postcode, str)`; `len(postcode) >= 5` | Pass | 0.000s |
| Generate field value decimal | `isinstance(val, (int, float))` | Pass | 0.174s |
| Generate field value text | `isinstance(val, str)` | Pass | 0.087s |
| Coordinates in thames range | `51.0 <= loc["lat"] <= 52.0`; `-1.0 <= loc["lon"] <= 1.0` | Pass | 2.638s |
| Processing stats | `stats["successful_properties"] == 5`; `stats["failed_properties"] == 0` | Pass | 1.326s |
| Output file exists | `result["file_path"].exists()` | Pass | 1.366s |
| Output file is valid json | `isinstance(data, dict)` | Pass | 1.298s |
| Generate correct count | `len(result["data"]["properties"]) == 5` | Pass | 1.350s |
| Generate returns expected keys | `"data" in result`; `"file_path" in result` (+1 more) | Pass | 1.383s |
| Property id format | `re.match(r"^PROP-[a-f0-9]{8}$", pid)` | Pass | 1.297s |
| Property ids unique | `len(ids) == len(set(ids))` | Pass | 2.579s |
| Construction year in range | `1600 <= year <= 2026` | Pass | 0.000s |
| Property id format | `re.match(r"^PROP-[a-f0-9]{8}$", meta["property_id"])` | Pass | 0.088s |
| Returns dict with required keys | `"property_id" in meta`; `"property_type" in meta` | Pass | 0.090s |
| Lines 73-75: generate_grid_reference returns a non-empty string. | `isinstance(ref, str)`; `len(ref) > 4` | Pass | 0.000s |
| Lines 50-52: generate_owner_name returns 'Firstname Lastname'. | `len(parts) == 2`; `all(p[0].isupper() for p in parts)` | Pass | 0.000s |
| Lines 43-45: generate_past_date returns YYYY-MM-DD string. | `isinstance(result, str)`; `len(parts) == 3` | Pass | 0.000s |
| Custom days_range is respected (result is a past date). | `dt < datetime.now()` | Pass | 0.000s |

## 1.1 Test Document History

| Date | Version | Description | Author |
|---|---|---|---|
| 05-Mar-2026 | 1.0 | Initial test suite (28 tests) | D. Kelly |
| 08-Mar-2026 | 1.1 | 25 test(s) added; 28 test(s) removed (total: 25) | D. Kelly |
| 08-Mar-2026 | 1.2 | 4 test(s) added (total: 29) | D. Kelly |
| 09-Mar-2026 | 1.3 | 12 test(s) added (total: 41) | D. Kelly |
| 21-Mar-2026 | 1.4 | 14 test(s) added (total: 55) | D. Kelly |